

233/2

CHEMISTRY PAPER 2

MARKING SCHEME

1 a)

- I) V
 II) S
 III) Q Neither gain nor lose electrons' hence only give more room for bond formation
 IV) Alkali metals
 V) On the grid
 VI) It ionizes by receiving an electron , hence the gained electrons repelled by the already existing electrons

b) i) SiCl_4 , PCl_5 , SCl_2 ,

Melting below room temperature and boiling point above room temperature

ii) AlCl_3 has simple molecular structure with weak van der waal forces , while MgCl_2 has giant ionic structure with strong ionic bond

2 a) To absorb excess carbon (iv) oxide or gas B

To absorb unreacted carbon (iv) oxide

ii) any carbonate /hydrogen carbonate and acid

b i) $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightarrow 2\text{CO}(\text{g})$ $\text{KOH}(\text{g}) + \text{CO}_2 \rightarrow \text{KHCO}_3(\text{aq})$

c) use of $\text{Ca}(\text{OH})_2$: CO does not form white precipitate with $\text{Ca}(\text{OH})_2$ while CO_2 does
 burning : CO burns with a blue flame while CO_2 does not support combustion

d) brown fumes produced

black substance dissolves

ii) $\text{C}(\text{s}) + 4\text{HNO}_3(\text{aq}) \rightarrow \text{CO}_2 + 4\text{NO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

e) concentrated sulphuric (iv) acid acts as dehydrating agent

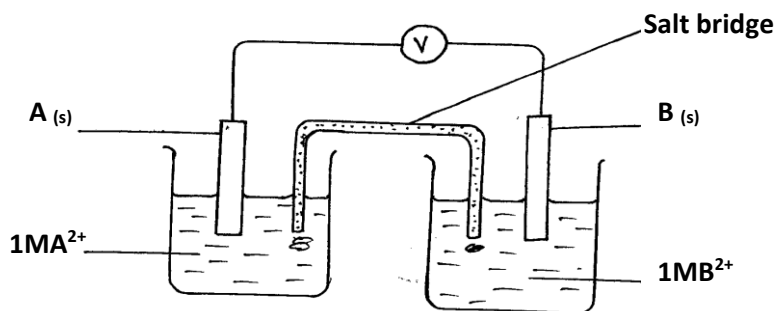
ii) - reducing agent in the extraction of metals from their ores

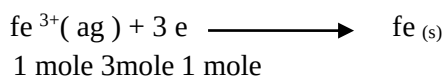
- used as fuels

- manufacture of hydrocarbons

3 a) (i) D^{+4} (iii) $\text{A}(\text{s})$

b i)

ii) $\text{emf} = E_{\text{red}} - E_{\text{ox}}$ - $.44 - (-0.776) = 0.32\text{v}$ iii) $\text{D}^{4+} / \text{D}^{3+}$ and $\text{A}_\text{s} / \text{A}^{2+}$



- 0.1 mole of Fe^{3+} + $0.1 \times 3 = 0.3$ faraday

d i) to prevent steel from rusting

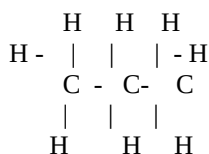
ii) to prevent steel from reacting with chromium at higher temperature since copper or nickel are unreactive

iii) $\text{Fe}^{2+}(\text{aq}) + \text{Cr}(\text{s}) \rightarrow \text{Cr}^{2+}(\text{aq}) + \text{Fe}(\text{s})$

iv) $Q = It = 4.5 \times 20 \times 60 \times 60 = 324,000 \text{ C}$

Mass deposited = $52 \times 3240000 / 96500 \times 2 = 87.295 \text{ g}$

4 a) propane



b) Reagent – concentrated sulphuric (vi) acid

Condition: $160 - 180^{\circ}\text{C}$

ii) condition: uv light /sunlight

c) (i) carbon(iv) oxide

(ii). hydrogen

(iii) propanoic acid

iv) Bromo-propan – 2-ol

2-bromopropan-1-ol

d) $2\text{C}_3\text{H}_7\text{OH} + 9\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 8\text{H}_2\text{O}(\text{g})$

e) Addition polymerization /polymerization

7) $3 \times 12 + 1 \times 6 = 42$

$N = 2100 / 42 = 500$

5a) phosphorus smolders

Its reaction with air is exothermic

II) phosphorous (iii)oxide formed is an acidic oxide which dissolves in water to form phosphoric acid.

iii) volume of air used = $30.65 - 24.28 = 6.37 \text{ cm}^3$

percentage volume = $6.37 / 30.65 \times 100 = 20.78\%$

b) black powder changes to red brown bead

brown bead Mg reduces CuO to copper metal

White ash formed : Mg is oxidized to MgO

C) carbon (iv) oxide

Sulphuric (iv) oxide

Nitrogen (iv) oxide

6) mass of magnesium = $20.36 - 19.52 = 0.84 \text{ g}$

Mass of oxygen = $20.92 - 20.36 = 0.56 \text{ g}$

Mass of MgO = $20.92 - 19.52 = 1.40 \text{ g}$

% of magnesium = $0.84 / 1.40 \times 100 = 60\%$

% of mass of oxygen MgO = $0.56 / 1.40 \times 100 = 40\%$

II) elements present Mg O



Mass	0.84	0.56
moles present	0.84/24	0.56/16
0.035	0.035	
Simple ratio		
Formula	MgO	



ii) I : moles ratio of NaOH ; $\text{H}_2\text{SO}_4 = 2:1$

Moles of NaOH = $2 \times 0.005 = 0.01$

ii) if 50 cm^3 of NaOH = 0.01

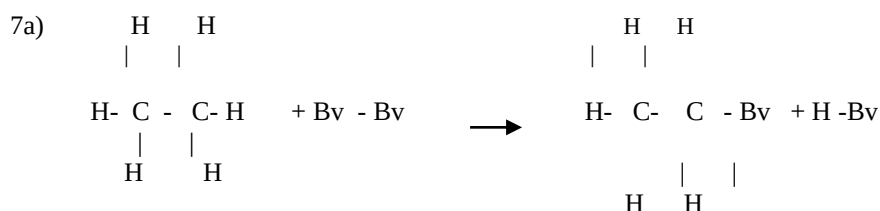
1000 cm^3 of NaOH = $1000 \times 0.01/50 = 0.02 \text{ moles}$

OR Ans in (i) $\times 1000/50 = \text{Ans}$

iii) molar mass of NaOH = $23 + 1 = 16 = 40 \text{ G}$

Mass of NaOH reacted = $40 \times 0.2 = 8 \text{ g}$

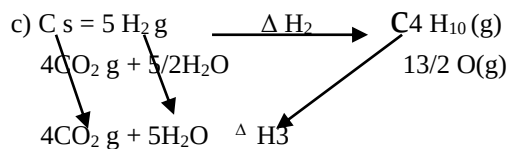
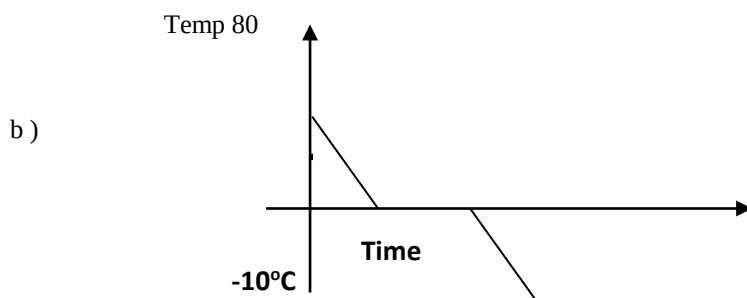
Mass of NaCl = $0.8/8.8 \times 100 = 9.0909\%$



Bond breaking energy = $413 + 193 = 606 \text{ kJ}$

Bond function energy = $280 + 365 = 645 \text{ kJ}$

ΔH for the reaction = $606 - 645$
= 39 kJ



$\Delta H_2 = \Delta H_1 - \Delta H_3$
 $= (4 \times -343 + -286) - (-2877)$
 $= -1572 - 1430 + 2877$
 $= -125 \text{ kJ/mol}$

d) $\Delta H + \Delta H_{\text{hyd}}$
 $= 690 + 322 + 364$
 $= 690 - 686$
 $= +4 \text{ kJ/mol}$